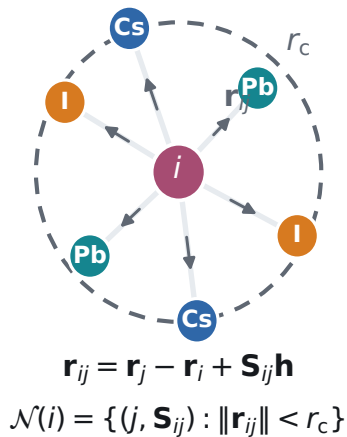
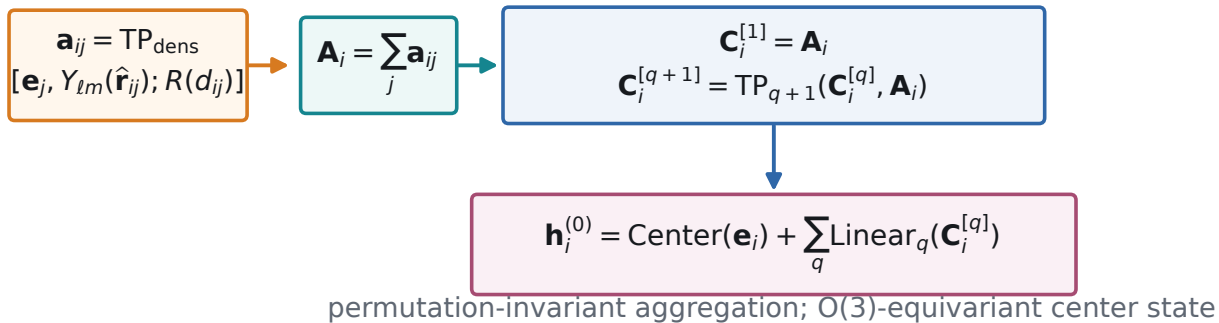


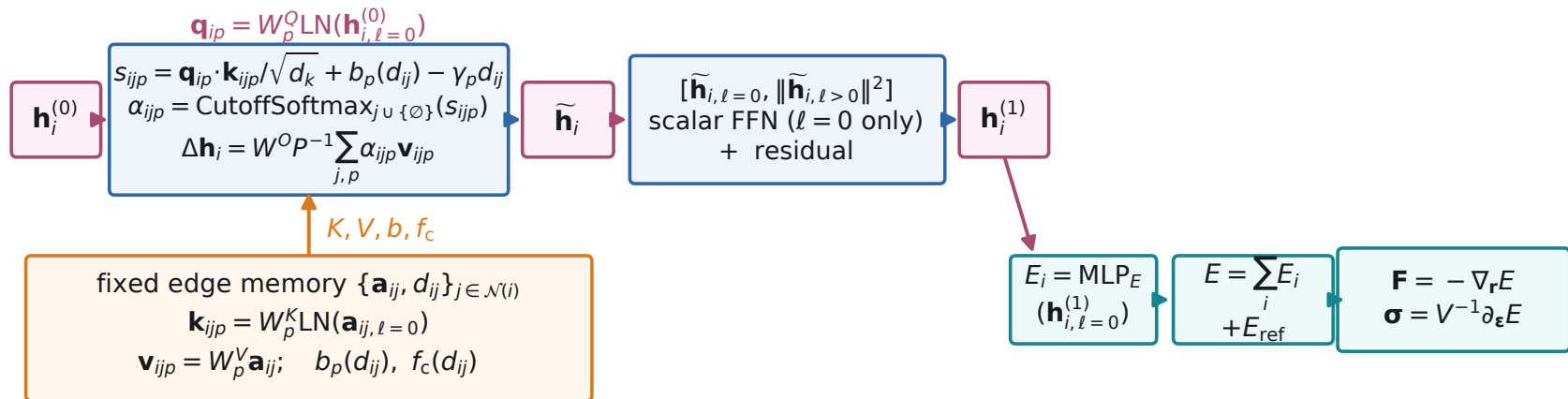
(a) Periodic environment



(b) ACE environment encoder



(c) One fixed-environment transformer block



All reported models use one block: only $\mathbf{h}_i$ is updated; $\mathbf{a}_{ij}$ remains fixed by species and geometry.